Giancarlo Frison

Senior AI Engineer | NLP, Neuro-symbolic AI, Enterprise Software.

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EU citizen

Hands-on software engineer with 20+ years of experience building production systems in enterprise and internet-scale environments. Strong background in NLP, hybrid symbolic/ML approaches, backend architecture. Focused on shipping reliable, maintainable systems for complex domains such as ERP/CRM and commerce.

AI Engineering: NLP, hybrid symbolic/deep learning systems, recommender systems.

Backend Engineering: distributed systems, event-driven high-throughput services, API design.

Data and Knowledge: knowledge graphs, query systems, declarative languages, workflows.

Delivery: prototyping, code quality, performance tuning, architecture implementation.

Collaboration: cross-functional engineering work, mentoring through code and design reviews.

SAP, Munich - *Software Research Engineer*

2017 - PRESENT

Key Projects & Innovations:

- **Hybrid AI for ERP Q&A**: Designed methods and implemented software ecosystem for answering natural language questions over a variety of information sources typical of an ERP/CRM system. It is an efficient and effective neuro-symbolic approach that minimizes customization and promotes separation of concerns between model based and deterministic modules with unpaired precision and granularity.
- **Declarative Optimization Language**: Designed and implemented a functional-logic programming system integrating relational algebra and combinatorial primitives, enabling efficient resolution of complex optimizations through a simple syntax humans and AI agents can easily generate and maintain.
- **Knowledge-Driven Recommender Engine**: Built a dynamic recommendation system leveraging knowledge graphs to generate logically justified product suggestions. Incorporated interactive preference elicitation to refine results and reasoning.
- **AI-Powered Sales Automation**: Engineered an adaptive sales assistant using Bayesian networks and reinforcement learning. The system personalizes offers and sales strategies by inferring customer preferences and behavioral traits through interactive dialogues.
- **Conversational Commerce**: Pioneered a neural network architecture for e-commerce interactions via Facebook Messenger and SAP Commerce Cloud. Enabled natural language product searches, order placement, FAQ navigation, and ticket creation.

SAP, Munich - *Development Architect*

2014 - 2017

- Designed and developed an inter-process software library for **SAP Commerce Cloud**, enabling seamless remote service access for developers. This standardized solution, now part of the SAP Commerce Toolkit, utilized **event-driven programming** to optimize resource efficiency.
- Enhanced the core platform's service layer and **ORM framework**, improving performance and scalability.
- Optimized **high-throughput** libraries within the SAP Commerce Toolkit to maximize efficiency and reduce latency.

TechPeaks, Trento - *Selected Attendee*

2013 - 2014

- Chosen for a competitive training program focused on **Lean Startup** methodologies, **business planning**, and corporate administration.
- Developed and pitched a **comprehensive business plan** for securing (successfully) financial grants.
- Strategic partnerships with key stakeholders to drive business growth.
- Obtained a food management business license, ensuring regulatory compliance.

Mérieux NutriSciences Chelab, Italy - *Development Architect*

2011 - 2013

- Conducted stakeholder interviews and analyzed business processes to assess software architecture needs, leading to the prototyping of the company's Laboratory Information Management System (LIMS).
- Designed a domain-specific language (DSL) to optimize chemical workflow management.
- Mentored the development team in rapid web application development using Grails.
- Established a delivery pipeline to streamline development and deployment processes.

Skebby, Milan - *Senior Software Developer*

2010 - 2011

- Designed and developed a distributed microservices architecture for high-volume B2C and B2B SMS delivery management.
- Mentored team members on best practices for scalable and resilient system design.
- Implemented an Enterprise Integration Pattern (EIP)-based architecture, leveraging ActiveMQ for inter-service communication.
- Optimized system performance and reliability through microservices orchestration and message-driven workflows.

Quinary, Milan - *Senior Software Developer*

2008 - 2010

- Re-engineering the SMS B2B delivery system for Vodafone based on micro-service architecture with messages broker (FioranoMQ).

Nimbuzz, The Netherland - *Senior Software Developer*

2007 - 2008

- Implemented high-throughput XMPP server-to-server Java framework for chat protocols through REST API.

Ats, Milan - *Senior Software Developer*

2006 - 2007

- Created the BrokerTec financial market extension, integrated into the trading application, enabling trades of REPO market government securities.
- Designed and implemented internal framework for interfacing external API in a declarative way, solely by XML definitions.

Abla, Milan - *Senior Software Developer*

2006 - 2007

- Developed the BrokerTec financial market extension, seamlessly integrating it into the trading platform to facilitate REPO market government securities transactions.
- Designed and implemented an internal framework to streamline external API interactions using a declarative XML-based approach, eliminating the need for manual coding.

SecServizi, Padova - *Software Developer*

2002 - 2006

- Developed custom extensions for IBM Rational Software Architect to support Low-Code/No-Code Model-Driven Architecture (MDA).
- Empowered business analysts and non-developers to design banking services through intuitive UML-based modeling tools.
- Streamlined software development processes, reducing dependency on manual coding for critical financial applications.

EDUCATION

Open University Cyprus - *Master Science - Cognitive Systems*

2023 - 2026

Master Thesis: *Programming Language and System for Enhancing AI-assisted Software Development*

Università degli studi di Padova - *Bachelor Eng - Ing. Informatica ed Automatica*

1995 - 1999

Istituto Tecnico Industriale - *High school - Diploma Perito Elettrotecnico*

1990 - 1995

PATENTS

Natural language & data programming for combinatorics - *19/365,428 Pending*
2025

Hypergraph store and query mechanism - *18/972,932 Pending*
2024

Scarlatti: In-memory hypergraph machine learning - *18/492,418 Granted*
2023

Time-based query processing for system dynamics - *18/078,558 Granted*
2023

Low-code algorithm for knowledge reasoning - *18/078,566 Pending*
2022

Consumer recommender with Knowledge Graphs - *EP4181030B1 Granted*
2021

GILP: Symbolic machine learning over graphs - *17/546,450 Granted*
2021

Improve Style in Commerce with Knowledge Graphs - *16/687,982 Granted*
2019

Foraging Preferences for E-Commerce Recommendation - *16/687,951 Granted*
2019

Smoothing conversational contexts for digital agents - *16/177,983 Granted*
2018

Concept search in e-commerce via word embeddings - *16/201,737 Granted*
2018

CONFERENCES

PX26 - *Speaker*

2026

Pull Down Programming Complexity with Kubrick

AI tools are now the horsepower of computer programming. They are generally great for writing glue-code and integration tasks, probably less than ideal for complex problems on complicated programming settings. What could be the reasons that prevent generators on full-scale adoption?

I am implementing a declarative programming language that facilitates the synergy between automatons and humans in software development by forcing AI tools to generate intuitive code and to allow human operators to understand what is in there. It is an attempt to lower the barriers by simplifying the programming experience.

In the presentation, I will touch on various aspects that include cognitive aspects of problem-solving applied to programming and the role of AI tools such as LRMs on software development. I will provide arguments for stating that the accidental complexity of the programming system may worsen the performance of AI generators as well as it affects human developers.

SEMANTiCS - *Speaker*

2022

The Socratic Digital Agent: persuasion in eCommerce with knowledge, logic, and machine learning

E-Stores lose sales due to the negative biases of consumers. While salespeople give proper reasons to change consumers' misbeliefs, it is problematic to address those issues in an online shop. In this talk, I will present how to combine semantic graphs with logic programming and symbolic machine learning to restore consumer confidence. The digital agent detects what problems users might have and offers them explanations and valid arguments for not worrying about, or why a given recommendation is more suitable than others.

CERTIFICATIONS

SAP Architect Curriculum

2022

Professional Scrum Master

2021

ITIL® Foundation Certificate in IT Service Management

2021